

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct:

I M. PRASANNA VENKATESH (192573016) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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BP

1) Smart Medical Diagnosis

a) Proportional Logic

* F = Fever

* C = Cough

* B = Breathlessness

* FI = Possible Flu

* M = Possible Measles

* P = Possible Pneumonia

Rules,

$$1. F \wedge C \rightarrow FI$$

$$2. F \wedge \text{Rash} \rightarrow M$$

$$3. C \wedge B \rightarrow P$$

Facts: F, C, B

b) Forward Chaining

$$1. F \wedge C \rightarrow FI$$

$$2. C \wedge B \rightarrow P$$

$$3. F \wedge \text{Rash} \rightarrow M \rightarrow \text{Cannot derive}$$

Diagnosis: Possible Flu, Possible Pneumonia.

c) Backward Chaining

Goal: P

$$P \leftarrow C \wedge B$$

$$C = \text{True}, B = \text{True}$$

$$\therefore P = \text{True},$$

2) AUTONOMOUS TRAFFIC MANAGEMENT

a) Unification

Rule 1:

$$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$$

Facts:

$$\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$$

Substitution:

$$\theta = \{x/\text{Ambulance}\}$$

$$\text{Therefore: } \text{ClearPath}(\text{Ambulance}).$$

b) Resolution

CNF:

1. $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
2. $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
3. $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$
4. $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}$

Facts:

$$\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{Car A}),$$

$$\text{Behind}(\text{Car A}, \text{Ambulance})$$

Form 1:

$$\text{ClearPath}(\text{Ambulance})$$

Form 2:

$$\text{GreenSignal}(\text{Ambulance})$$

Using Rule 4:

Vehicle(Car A) + Behind(Car A, Ambulance) + Green Signal(Ambulance)

$\therefore$ Proceed(Car A)

c) Two Emergency Vehicles

The KB is partially sufficient, because Rule 1 applies to every emergency vehicle. But it needs facts such as
Vehicle(Amb2), EmergencyType(Amb2)

3) Agricultural Expert System

a) CNF

P1: Soil Dry $\rightarrow$ Irrigation Needed

$= \neg \text{Soil Dry} \vee \text{Irrigation Needed}$

P2: Irrigation Needed $\wedge$ Grow Wheat $\rightarrow$ Apply Drip Method

$= \neg \text{Irrigation Needed} \vee \neg \text{Grow Wheat} \vee \text{Apply Drip Method}$

P3: $\neg \text{Apply Drip Method} \wedge \text{Grow Wheat} \rightarrow \text{Grow At Risk}$

$= \text{Apply Drip Method} \vee \neg \text{Grow Wheat} \vee \text{Grow At Risk}$

Facts: Soil Dry, Grow Wheat

b) Resolution Refutation

Goal: Apply Drip Method

Negate: $\neg \text{Apply Drip Method}$

1. Soil Dry + $P_1 \rightarrow$ Irrigation Needed

2. Irrigation Needed + Crop At Risk + $P_2 \rightarrow$ Apply Drip Method

3. Apply Drip Method + $\neg$ Apply Drip Method $\rightarrow \perp$

Therefore, Apply Drip Method is proven.

Soil Dry $\rightarrow$ Irrigation Needed $\rightarrow$ Crop At Risk $\rightarrow$ Apply Drip Method.

c) Remove Soil Dry

Without Soil Dry:

- * Cannot derive Irrigation Needed

- * Cannot derive Apply Drip Method

- * Cannot derive Crop At Risk, because its rule requires $\rightarrow$ Apply Drip Method, which is also not known.

Conclusion: Crop At Risk does not follow.

4) Mumbaz World

a) Belief State

Percepts:

- * Stewch [1, 2]

- * Breeze [1, 1]

- * Glitter [2, 2]

Using Modus Ponens:

1. Stewch [1, 2] $\rightarrow$ Mumbaz adjacent to [1, 2]

2. Breeze [1, 1] $\rightarrow$ Pit adjacent to [1, 1]

3. Clutter $[2,2] \rightarrow$ Gold at $[2,2]$

b) Possible Locations

* Mumpus: adjacent to $[1,2] \rightarrow [1,1], [1,3], [2,2]$

* Pit: adjacent to $[1,1] \rightarrow [1,2], [2,1]$

c) Path Safety

Path:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

* $[1,2]$ may contain a Pit

* $[2,2]$ may contain a Mumpus

* Therefore, the path is Not guaranteed safe.

Conclusion:

Gold is at $[2,2]$, but the agent should not assume the path is safe.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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AA